

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412424
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Karthik; Gayathri et al.

Predicting avatars for a user based on emotions of the user

Abstract

A system receives a first image and extracts a first set of facial features from the first image. The system determines that a user exhibited a first emotion in the first image. The system determines that the user selected a first avatar for use within a virtual environment. The system associates the first avatar with the first emotion of the user. The system receives a second image and extracts a second set of facial features from the second image. The system determines that the user exhibited a second emotion in the second image. The system determines that the first set of facial features corresponds to the second set of facial features. In response, the system presents the first avatar to the user. The system determines that the user selects the first avatar. The system implements the first avatar on behalf of the user for use in the virtual environment.

Inventors: Karthik; Gayathri (Tamil Nadu, IN), Mohan; Kumaraguru (Tamil Nadu, IN), Jaiswal; Vivek (Telangana, IN), M M; Logesh (Tamil Nadu, IN), Babu; Vasantha Kumar (Tamil Nadu, IN)

Applicant: Bank of America Corporation (Charlotte, NC)

Family ID: 92935240

Assignee: Bank of America Corporation (Charlotte, NC)

Appl. No.: 18/296600

Filed: April 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240338972 A1	Oct. 10, 2024

Publication Classification

Int. Cl.: G06V40/16 (20220101); G06V10/75 (20220101); G06V10/774 (20220101)

U.S. Cl.:

CPC **G06V40/174** (20220101); **G06V10/751** (20220101); **G06V10/774** (20220101);
G06V40/168 (20220101);

Field of Classification Search

CPC: G06V (40/174); G06V (40/168); G06V (10/751); G06V (10/774); G06V (40/171); G06V (40/176)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8063929	12/2010	Kurtz et al.	N/A	N/A
8112490	12/2011	Upton et al.	N/A	N/A
8149241	12/2011	Do et al.	N/A	N/A
8154578	12/2011	Kurtz et al.	N/A	N/A
8154583	12/2011	Kurtz et al.	N/A	N/A
8159519	12/2011	Kurtz et al.	N/A	N/A
8237771	12/2011	Kurtz et al.	N/A	N/A
8253770	12/2011	Kurtz et al.	N/A	N/A
8274544	12/2011	Kurtz et al.	N/A	N/A
9808722	12/2016	Kawachiya et al.	N/A	N/A
9864742	12/2017	Chen et al.	N/A	N/A
10326667	12/2018	Jones et al.	N/A	N/A
2009/0177969	12/2008	Jones et al.	N/A	N/A
2017/0259167	12/2016	Cook et al.	N/A	N/A
2018/0091732	12/2017	Wilson	N/A	G06F 3/048
2019/0250934	12/2018	Kim	N/A	H04M 1/72436
2019/0266775	12/2018	Lee	N/A	G06F 3/0346
2020/0193667	12/2019	Chu	N/A	G06T 11/60
2020/0234481	12/2019	Scapel	N/A	A63F 13/213
2021/0357447	12/2020	Mckenzie	N/A	N/A
2022/0198254	12/2021	Dalli et al.	N/A	N/A
2022/0240779	12/2021	Peyman	N/A	N/A
2022/0248955	12/2021	Tran	N/A	N/A
2022/0318852	12/2021	Blaikie, III et al.	N/A	N/A
2023/0215070	12/2022	Jordan	N/A	G06T 13/205

OTHER PUBLICATIONS

Jia Zheng Lim et al., "Emotion Recognition Using Eye-Tracking: Taxonomy, Review and Current Challenges," MDPI, Sensors 2020, 20, 2384; doi:10.3390/s20082384,
www.mdpi.com/journal/sensors. cited by applicant

Primary Examiner: Haider; Syed

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to data processing, and more specifically to predicting avatars for a user based on emotions of the user.

BACKGROUND

(2) Users of virtual environments may use avatars to operate within the virtual environment. Users are provided a wide variety of avatars to choose from when they want to access the virtual environment. Most if not all of the generated avatars are generic and may not be of interest to the users. A large number of processing and memory resources are spent to generate these avatars. Similarly, a large number of network resources are spent to communicate these avatars to user devices.

SUMMARY

(3) The system described in the present disclosure is particularly integrated into practical applications of improving the processing, memory, and network allocation and utilization for generating avatars for users to operate within virtual environments, and communicating the generated avatars to the users for user selection. These practical applications and technical advantages that are derived from them are described below.

(4) The disclosed system is configured to reduce processing, memory, and network resources that would be otherwise spent to generate and present avatars that are irrelevant to the user (and user's emotion) using the current virtual environment management technology. In the current virtual environment management technology, users are presented with a wide variety of avatars to choose from when they want to operate within the virtual environment. A large number of processing and memory resources are spent to generate and maintain these wide varieties of avatars in databases. Most of the generated avatars are irrelevant and not of interest to the users. Therefore, most of the generated avatars are never selected by the users. Thus, the spent processing and memory resources are wasted. Furthermore, network resources are used to communicate the generated avatars to the users on their user devices and/or on an application platform where the users can access the virtual environments-most of these generated avatars turn out to be irrelevant to the user and to the user's emotion at the time of the avatar selection. Thus, the spent network resources are wasted using the current virtual environment management technology.

(5) Certain embodiments of the disclosed system are configured to provide a technical solution to the above-identified and other technical problems arising in the realm of virtual environment technology. The present disclosure is configured to contemplate unconventional systems and methods to predict the emotion of the user based on images of the user, determine which avatars the user had selected previously when they had the same or similar emotions, associate or link the selected avatars to the detected emotions of the user at the time of the avatar selection, provide a list of recommended avatars that are associated with the same or similar emotions to the user, and implement an avatar that the user selects among the list of recommended avatars on behalf of the user to operate within the virtual environment. Therefore, the system is configured to learn the association and link between a selected avatar and the emotion of the user at the time of selection of the avatar **104**.

(6) In certain embodiments, the disclosed system is configured to generate more relevant avatars that have one or more combined attributes of avatars for each emotion of the user and present the generated avatars along with the historically selected avatars per each case of detected emotion of the user. This improves the efficiency in the utilization of processing, memory, and network resources that are typically spent on generating and presenting irrelevant avatars using the current techniques.

(7) In this manner, the disclosed system provides the practical application of conserving processing, memory, and network resources that would otherwise be spent on generating and presenting irrelevant avatars to the user using the current technology. For example, by classifying the emotions of the user with respective avatars that the user historically elected, the disclosed system is configured to generate and present more relevant avatars per each emotion of the user. Thus, the processing, memory, and network resources are allocated more efficiently to the operations of generating and presenting the more relevant avatars per each emotion of users.

(8) In certain embodiments, a system for predicting avatars for a user based on the emotion of the user comprises a camera and a processor. The camera is configured to capture a set of images of a user, wherein the set of images comprises a first image and a second image. The processor is operably coupled to the camera. The processor is configured to receive the first image from the camera. The processor is further configured to extract a first set of facial features from the first image. The processor is further configured to determine that the user exhibited a first emotion in the first image based at least in part upon the first set of extracted facial features. The processor is further configured to determine that the user selects a first avatar for use within a virtual environment. The processor is further configured to associate the first emotion of the user to the first avatar. The processor is further configured to build a training dataset that comprises the first emotion of the user labeled with the first avatar. The processor is further configured to receive the second image from the camera. The processor is further configured to extract a second set of facial features from the second image. The processor is further configured to determine that the user exhibited a second emotion in the second image based at least in part upon the second set of extracted facial features. The processor is further configured to compare at least one of the first set of extracted facial features to a counterpart facial feature from among the second set of extracted facial features. The processor is further configured to determine that more than a threshold percentage of the first set of extracted facial features correspond to the counterpart facial features from among the second set of extracted facial features. In response to determining that more than the threshold percentage of the first set of extracted facial features correspond to the counterpart facial features from among the second set of extracted facial features, the processor is further configured to determine that the first emotion of the user corresponds to the second emotion of the user exhibited in the second image. The processor is further configured to present at least the first avatar that is labeled with the first emotion for user selection. The processor is further configured to implement the first avatar on behalf of the user in the virtual environment.

(9) Certain embodiments of this disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 illustrates an embodiment of a system configured to predict avatars for a user based on emotions of the user;

(3) FIG. 2 illustrates a block diagram of an embodiment of a computing device used by the system **100** of FIG. 1;

(4) FIG. 3 illustrates an example operational flow of the system of FIG. 1 to predict avatars for a user based on emotions of the user; and

(5) FIG. 4 illustrates an example flowchart of a method to predict avatars for a user based on

emotions of the user.

DETAILED DESCRIPTION

(6) As described above, previous technologies fail to provide efficient and reliable solutions to predict avatars for a user based on the emotions of the user. Embodiments of the present disclosure and its advantages may be understood by referring to FIGS. 1 through 4. FIGS. 1 through 4 are used to describe systems and methods that predict avatars for a user based on the emotions of the user.

(7) System Overview

(8) FIG. 1 illustrates an embodiment of a system **100** that is generally configured to i) predict avatars **104** for a user **102** based on the emotion **156** of the user **102** to operate within the virtual environment **130** and ii) predict virtual locations **132** for the user **102** based on the emotion **156** of the user **102** to visit within the virtual environment **130**. In certain embodiments, the system **100** comprises a computing device **120** communicatively coupled to the virtual environment management device **140** via a network **110**. Network **110** enables communications among the components of the system **100**. The computing device **120** includes a processor (**202** in FIG. 2) in signal communication with a memory (**220** in FIG. 2). Memory (**220** in FIG. 2) stores software instructions (**222** in FIG. 2) that when executed by the processor (**202** in FIG. 2) cause the processor to perform one or more operations described herein. The virtual environment management device **140** includes a processor **142** in signal communication with a memory **146**. Memory **146** stores software instructions **148** that when executed by the processor **142** cause the processor **142** to perform one or more operations described herein. In other embodiments, system **100** may not have all of the components listed and/or may have other elements instead of, or in addition to, those listed above.

(9) In general, the system **100** reduces processing, memory, and network resources that would be otherwise spent to generate and present avatars **104** that are irrelevant to the user **102** (and user's emotion **156**) using the current virtual environment management technology. In the current virtual environment management technology, the users **102** are presented with a wide variety of avatars **104** to choose from when they want to operate within the virtual environment **130**. A large number of processing and memory resources are spent to generate these wide varieties of avatars **104**. Most of the generated avatars **104** may be irrelevant and not of interest to the users **102**. Thus, the spent processing and memory resources are wasted. Furthermore, network resources are used to communicate the generated avatars **104** to the users **102** which may turn out to be irrelevant to the user **102** and to the user's emotion at the time of the avatar selection. Thus, the spent network resources are wasted using the current virtual environment management technology.

(10) Certain embodiments of the system **100** are configured to provide a solution to this and other technical problems arising in the realm of virtual environment management technology. The present disclosure is configured to contemplate unconventional systems and methods to predict the emotion of the user **102** based on images of the user **102**, determine which avatars **104** the user **102** had selected previously when they had the same or similar emotions, associate or link the selected avatars **104** to the detected emotions **156** of the user **102** at the time of the avatar selection, provide a list of recommended avatars **104** that are associated with the same or similar emotions to the user **102**, and implement an avatar **104** that the user selects among the list of recommended avatars **104** on behalf of the user **102** to operate within the virtual environment **130**. Therefore, the system **100** is configured to learn the association and link between a selected avatar **104** and the emotion of the user **102** at the time of selection of the avatar **104**.

(11) In certain embodiments, the system **100** is configured to generate more relevant avatars **104** that have combined attributes of avatars **104** for each emotion of the user **102** and present the generated avatars **104** along with the historically selected avatars **104** per each case of detected emotion of the user **102**. This improves the efficiency in utilization of processing, memory, and network resources that are typically spent for generating and presenting irrelevant avatars **104**

using the current technology.

(12) In this manner, the system **100** provides the practical application of conserving processing, memory, and network resources that would otherwise be spent on generating and presenting irrelevant avatars **104** to the user **102** using the current technology. For example, by classifying emotions **156** of the user **102** with respective avatars **104** that the user **102** historically elected, the system **100** is configured to generate and present more relevant avatars **104** per each emotion **156** of the user **102**. Thus, the processing, memory, and network resources are allocated more efficiently to operations of generating and presenting the more relevant avatars **104** per each emotion **156** of users **102**.

(13) System Components

(14) Network

(15) Network **110** may be any suitable type of wireless and/or wired network. The network **110** may be connected to the Internet or public network. The network **110** may include all or a portion of an Intranet, a peer-to-peer network, a switched telephone network, a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), a wireless PAN (WPAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a plain old telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMAX, etc.), a long-term evolution (LTE) network, a universal mobile telecommunications system (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a near-field communication (NFC) network, and/or any other suitable network. The network **110** may be configured to support any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(16) Computing Device

(17) Computing device **120** is described in detail in the discussion of FIG. 2. In brief, a computing device **120** is generally any device that is configured to process data and interact with users **102**. Examples of the computing device **120** include, but are not limited to, a personal computer, a desktop computer, a workstation, a server, a laptop, a tablet computer, a mobile phone (such as a smartphone), smart glasses, Virtual Reality (VR) glasses, a virtual reality device, an augmented reality device, an Internet-of-Things (IoT) device, or any other suitable type of device. The computing device **120** may include a user interface, such as a display, a microphone, a camera, keypad, or other appropriate terminal equipment usable by user **102**. The computing device **120** may include a hardware processor, memory, and/or circuitry (see FIG. 2) configured to perform any of the functions or actions of the computing device **120** described herein. For example, a software application designed using software code may be stored in the memory and executed by the processor to perform the functions of the computing device **120**. The computing device **120** is configured to communicate with other devices and components of the system **100** via the network **110**. In the present disclosure, the computing device **120** may be interchangeably referred to as a user device.

(18) FIG. 2 illustrates a block diagram of an embodiment of the computing device **120** used by the system **100** of FIG. 1. The computing device **120** may be configured to display the virtual environment **130** (see FIG. 1) within a field of view of the user (see FIG. 1), capture biometric, sensory, and/or physical information of the user wearing and/or operating the computing device **120**, and to facilitate an electronic interaction between the user and other components of the system **100** (see FIG. 1), including the virtual environment management device **140**.

(19) In the illustrated embodiment, the computing device **120** comprises a processor **202**, a memory **220**, a display **206**, a camera **208**, a wireless communication interface **210**, a network interface **212**, a microphone **214**, a global positioning system (GPS) sensor **216**, and biometric devices **218**. The components of the computing device **120** are operably coupled to each other via wires and/or wireless communications. The processor **202** comprises one or more processors operably coupled to and in signal communication with memory **220**, display **206**, camera **208**,

wireless communication interface **210**, network interface **212**, microphone **214**, GPS sensor **216**, and biometric devices **218**. The one or more processors is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **202** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **202** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **202** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components. The one or more processors are configured to implement various instructions, including software instructions that when executed by the processor **202** cause the computing device **120** to perform one or more operations of the computing device **120** described herein. For example, the one or more processors are configured to execute instructions to implement the function disclosed herein, such as some or all of those described with respect to FIGS. **1-4**. For example, processor **202** may be configured to display the virtual environment (**130** in FIG. **1**), virtual objects and locations within the virtual environment (**130** in FIG. **1**) on display **206**, detect avatar location in the virtual environment, detect user location in the real world, identify virtual sub-environment/locations within the virtual environment, capture biometric information of a user, via one or more of camera **208**, microphone **214**, and/or biometric devices **218**, and communicate via wireless communication interface **210** with virtual environment management device **140** (see FIG. **1**) and/or other user devices **120** (see FIG. **1**).

(20) Memory **220** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **220** may include one or more of a local database, a cloud database, a network-attached storage (NAS), etc. The memory **220** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **220** may store any of the information described in FIGS. **1-4** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **202**. For example, the memory **220** may store software instructions **222**, emotion recognition machine learning algorithm **224**, images **226a-c**, sensor data **228a-c**, training dataset **154a**, vectors **312a, b**, facial features **310a, b**, avatar prediction machine learning algorithm **150**, virtual location prediction machine learning algorithm **152**, training dataset **154b**, emotion data **156a,b**, and/or any other data or instructions. The software instructions **222** may comprise any suitable set of instructions, logic, rules, or code operable to execute the processor **202** and perform the functions described herein, such as some or all of those described in FIGS. **1-4**. In some embodiments, the memory **220** may store and/or include a browser or web application for a computing device **120** to interact with the virtual locations in the virtual environment **130** and other devices.

(21) The emotion recognition machine learning algorithm **224** may be implemented by the processor **202** executing software instructions **222** and is generally configured to detect emotions of the user based on images **226** (e.g., images **226a-c**) and/or sensor data **228** (e.g., sensor data **228a-c**). In certain embodiments, the emotion recognition machine learning algorithm **224** may include a support vector machine, neural network, random forest, k-means clustering, Tree-based algorithm, Random Forest algorithm, etc. In certain embodiments, the emotion recognition machine learning algorithm **224** may be implemented by a plurality of neural network layers, convolutional neural

network layers, Long-Short-Term-Memory (LSTM) layers, Bi-directional LSTM layers, recurrent neural network layers, and the like. In certain embodiments, the emotion recognition machine learning algorithm **224** may include a neural network image processing algorithm.

(22) The emotion recognition machine learning algorithm **224** may be configured to be implemented by unsupervised, semi-supervised, and/or supervised machine learning algorithm. For example, the emotion recognition machine learning algorithm **224** may be pre-trained to determine the emotion **156** of the user **102** from the image **226** based on a training dataset that comprises a set of image samples, each labeled with a respective emotion **156**. The emotion recognition machine learning algorithm **224** may be given a training image labeled with a respective emotion **156**. The emotion recognition machine learning algorithm **224** may extract a set of facial features from the training image. The emotion recognition machine learning algorithm **224** may learn to associate the extracted set of facial features to the emotion label. The emotion recognition machine learning algorithm **224** may perform this operation for multiple training images to build the training dataset.

(23) In the testing phase, the emotion recognition machine learning algorithm **224** may be given a testing image and asked to predict the emotion of a user shown in the testing image. The emotion recognition machine learning algorithm **224** may extract a set of facial features from the testing image. The emotion recognition machine learning algorithm **224** may compare the extracted set of facial features to the facial features provided in the training dataset. The emotion recognition machine learning algorithm **224** may determine to which of the provided facial features in the training dataset, the newly extracted facial features correspond or match with. In response to finding a match, the emotion recognition machine learning algorithm **224** may determine that the newly extracted facial features are associated with the same emotion as the corresponding or matching facial features as provided in the training dataset. Similar operations may be performed with respect to the sensor data **228**. For example, the emotion recognition machine learning algorithm **224** may be pre-trained to determine the emotion **156** of the user **102** from the sensor data **228** based on a training dataset that comprises a set of sensor data samples, each labeled with a respective emotion **156**.

(24) Display **206** may be or include a display screen that allows displaying objects (e.g., virtual and/or physical objects). Display **206** is configured to present visual information to a user (for example, user **102** in FIG. **1**) in an augmented reality environment that overlays virtual or graphical objects onto tangible objects in a real scene in real-time. In certain embodiments, the display **206** is configured to present visual information to the user as the virtual environment **130** (see FIG. **1**) in real-time. In certain embodiments, display **206** is a wearable optical display (e.g., glasses or a headset) configured to reflect projected images and enables a user to see through the display **206**. For example, display **206** may comprise display units, lens, semi-transparent mirrors embedded in an eye glass structure, a visor structure, or a helmet structure. Examples of display units include, but are not limited to, a cathode ray tube (CRT) display, a liquid crystal display (LCD), a liquid crystal on silicon (LCOS) display, a light emitting diode (LED) display, an active-matrix OLED (AMOLED), an organic LED (OLED) display, a projector display, or any other suitable type of display as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. In certain embodiments, display **206** is a graphical display on a computing device **120**. For example, the graphical display may be the display of a tablet or smart phone configured to display an augmented reality environment with virtual or graphical objects overlaid onto tangible objects in a real scene in real-time environment and/or virtual environment **130**.

(25) Camera **208** may be or include one or more cameras that are configured to capture images **226** of a field of view in front of the computing device **120** and/or in front of the camera **208**. Examples of the camera **208** may include charge-coupled device (CCD) cameras and complementary metal-oxide semiconductor (CMOS) cameras. For example, camera **208** may be configured to capture images **226** of a wearer or user of the computing device **120**. Camera **208** may be configured to capture images continuously, at predetermined intervals, or on-demand. For example, camera **208**

may be configured to receive a command from the user to capture an image **226**. In another example, camera **208** is configured to continuously capture images to form a video stream. Camera **208** is communicably coupled to processor **202**.

(26) Wireless communication interface **210** may be or include any communication interface that allows communication with other devices. The wireless communication interface **210** is configured to facilitate processor **202** in communicating with other devices. Wireless communication interface **210** is configured to employ any suitable communication protocol, including WIFI, Bluetooth, among others.

(27) Network interface **212** is configured to enable wired and/or wireless communications. The network interface **212** may be configured to communicate data between the computing device **120** and other devices, systems, or domains of the system **100** (see FIG. **1**). For example, the network interface **212** may comprise a near field communication (NFC) interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, a radio-frequency identification (RFID) interface, a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a metropolitan area network (MAN) interface, a personal area network (PAN) interface, a wireless PAN (WPAN) interface, a modem, a switch, and/or a router. The processor **202** may be configured to send and receive data using the network interface **212**. The network interface **212** may be configured to use any suitable type of communication protocol.

(28) Microphone **214** may be or include a sound-detecting sensor or sound-detecting sensor array. Microphone **214** is generally configured to capture audio signals (e.g., voice signals or commands) from a user. Microphone **214** is communicably coupled to processor **202**. Microphone **214** may communicate the detected audio signals to the processor **202**.

(29) GPS sensor **216** may be configured to capture and provide geographical location information. For example, GPS sensor **216** is configured to provide a geographic location of a computing device **120**. GPS sensor **216** may be configured to provide the geographic location information as a relative geographic location or an absolute geographic location. GPS sensor **216** may provide the geographic location information using geographic coordinates (i.e., longitude and latitude) or any other suitable coordinate system. GPS sensor **216** is communicably coupled to processor **202**. GPS sensor **216** may communicate the captured location of the computing device **120** to the processor **202**.

(30) Examples of biometric devices **218** may include, but are not limited to, electrooculography sensors, retina scanners, fingerprint scanners, and the like. Biometric device **218** is communicably coupled to processor **202**. Biometric devices **218** are configured to capture sensor data **228** that includes information about a person's physical characteristics and to output a biometric signal based on captured information. The biometric device **218** may communicate the captured biometric features as sensor data **228** to the processor **202**.

(31) Referring back to FIG. **1**, the computing device **120** is configured to display a two-dimensional (2D) or three-dimensional (3D) representation of a virtual environment **130** to a user. Examples of a virtual environment **130** include, but are not limited to, a graphical or virtual representation of a metaverse, a map, a city, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment.

(32) Virtual Environment Management Device

(33) Virtual environment management device **140** generally includes a hardware computer system configured to manage activities within and with respect to the virtual environment **130**. In certain embodiments, the virtual environment management device **140** may be implemented by a cluster of computing devices, such as virtual machines. For example, the virtual environment management device **140** may be implemented by a plurality of computing devices using distributed computing and/or cloud computing systems in a network. In certain embodiments, the virtual environment management device **140** may be configured to provide services and resources (e.g., data and/or hardware resources) to the components of the system **100**. For example, the virtual environment

management device **140** may be configured to predict avatars **104** based on emotions **156** of the users **102** and communicate the recommended avatars **104** to the user **102**, e.g., via the computing device **120**. In another example, the virtual environment management device **140** may be configured to predict virtual locations **132** based on emotions **156** of the users **102** and communicate the recommended virtual locations **132** to the user **102**, e.g., via the computing device **120**.

(34) The virtual environment management device **140** comprises a processor **142** operably coupled with a network interface **144** and a memory **146**. Processor **142** comprises one or more processors operably coupled to the memory **146**. The processor **142** is any electronic circuitry, including, but not limited to, state machines, one or more CPU chips, logic units, cores (e.g., a multi-core processor), FPGAs, ASICs, or DSPs. For example, one or more processors may be implemented in cloud devices, servers, virtual machines, and the like. The processor **142** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable number and combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **142** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **142** may include an ALU for performing arithmetic and logic operations. The processor **142** may register the supply operands to the ALU and stores the results of ALU operations. The processor **142** may further include a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components. The one or more processors are configured to implement various software instructions. For example, the one or more processors are configured to execute instructions (e.g., software instructions **148**) to perform the operations of the virtual environment management device **140** described herein. In this way, processor **142** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **142** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **142** is configured to operate as described in FIGS. 1-4. For example, the processor **142** may be configured to perform one or more operations of the operational flow **300** described in FIG. 3 and one or more operations of the method **400** as described in FIG. 4.

(35) Network interface **144** is configured to enable wired and/or wireless communications. The network interface **144** may be configured to communicate data between the virtual environment management device **140** and other devices, systems, or domains of the system **100**. For example, the network interface **144** may comprise an NFC interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, an RFID interface, a WIFI interface, a LAN interface, a WAN interface, a MAN interface, a PAN interface, a WPAN interface, a modem, a switch, and/or a router. The processor **142** may be configured to send and receive data using the network interface **144**. The network interface **144** may be configured to use any suitable type of communication protocol.

(36) The memory **146** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **146** may include one or more of a local database, cloud database, network-attached storage (NAS), etc. The memory **146** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **146** may store any of the information described in FIGS. 1-4 along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **142**. For example, the memory **146** may store software instructions **148**, avatar prediction machine learning algorithm **150**, training dataset **154a**, messages **106**, training dataset **154b**, virtual location prediction machine learning algorithm **152**, and/or any other data or instructions. The software instructions **148** may comprise any suitable set of instructions, logic,

rules, or code operable to execute the processor **142** and perform the functions described herein, such as some or all of those described in FIGS. **1-4**. The messages **106** may include any information stored in the memory (**220** of FIG. **2**) of the computing device **120**. The computing device **120** may communicate the messages **106** to the virtual environment management device **140** at any appropriate time, e.g., before or when the user **102** wants to select an avatar **104** to operate within the virtual environment **130**.

(37) The avatar prediction machine learning algorithm **150** may be implemented by the processor **142** executing software instructions **148** and is generally configured to predict avatars **104** based on the emotion **156** of the user **102** (indicated in the message **106** received from the computing device **120**). In certain embodiments, the avatar prediction machine learning algorithm **150** may include a support vector machine, neural network, random forest, k-means clustering. Tree-based algorithm, Random Forest algorithm, etc. In certain embodiments, the avatar prediction machine learning algorithm **150** may be implemented by a plurality of neural network layers, convolutional neural network layers, Long-Short-Term-Memory (LSTM) layers, Bi-directional LSTM layers, recurrent neural network layers, and the like. The avatar prediction machine learning algorithm **150** may be configured to be implemented by unsupervised, semi-supervised, and/or supervised machine learning algorithm. For example, the avatar prediction machine learning algorithm **150** may be configured to build the training dataset **154a** that includes a set of user's emotions **156**, each labeled with one or more respective avatars **104**. The avatar prediction machine learning algorithm **150** may use the training dataset **154a** to determine the recommended avatars **104** based on the detected emotion **156** of the user **102**. This operation is described in greater detail in FIG. **3**.

(38) The virtual location prediction machine learning algorithm **152** may be implemented by the processor **142** executing software instructions **148** and is generally configured to predict virtual locations **132** based on the emotion **156** of the user **102** (indicated in the message **106** received from the computing device **120**). In certain embodiments, the virtual location prediction machine learning algorithm **152** may include a support vector machine, neural network, random forest, k-means clustering. Tree-based algorithm, Random Forest algorithm, etc. In certain embodiments, the virtual location prediction machine learning algorithm **152** may be implemented by a plurality of neural network layers, convolutional neural network layers, Long-Short-Term-Memory (LSTM) layers, Bi-directional LSTM layers, recurrent neural network layers, and the like. The virtual location prediction machine learning algorithm **152** may be configured to be implemented by unsupervised, semi-supervised, and/or supervised machine learning algorithm. For example, the virtual location prediction machine learning algorithm **152** may be configured to build the training dataset **154b** that includes a set of user's emotions **156**, each labeled with one or more respective virtual locations **132**. The virtual location prediction machine learning algorithm **152** may use the training dataset **154b** to determine the recommended virtual locations **132** based on the detected emotion **156** of the user **102**.

(39) Operational Flow for Predicting Avatars Based on User's Emotions

(40) FIG. **3** illustrates an example operational flow **300** of the system **100** (see FIG. **1**) for predicting avatars **104** based on user's emotions **156**. In an example scenario, assume that the user **102** wants to select an avatar **104** to operate within the virtual environment **130**. The user **102** may log into their profile with respect to the virtual environment **130** using their credentials. In a case where the user device **120** is VR glasses, the user **102** may put on the user device **120**.

(41) In operation, the user device **120** and/or the virtual environment management device **140** may first build up the training dataset **154a** that includes a set of user's emotions **156**, each labeled with one or more respective customized avatars **104**. This operation is described below.

(42) Training Phase to Train the Avatar Prediction Machine Learning Algorithm

(43) In the training phase **302**, the user device **120** may receive a first image **226a** of the user **102** from the camera **208**. For example, in the case that the camera **208** is external to the user device **120**, the camera **208** may communicate the image **226a** to the user device **120**. In another example,

in the case that the camera **208** is internal to the user device **120**, the camera **208** may communicate the image **226a** to the processor of the user device **120**. In response to the user **102** logging into their profile and access a platform to enter the virtual environment **130**, the camera **208** associated with the user device **120** may capture the image **226a** of the user **102**. Receiving the first image **226a** from the camera **208** may be in response to the user **102** being logged into their user profile with respect to the virtual environment **130** to access the virtual environment **130** at a first timestamp. Similarly, the user device **120** may receive sensor data **228a** from the biometric device **218**. For example, the sensor data **228a** may include the average of the potential voltage signals between front and back of the eye of the user **102**, eye movements, and eye tracking information, among other information during the time window that the user **102** puts on the user device **120**.

(44) The user device **120** may feed the image **226a** and the sensor data **228a** to the emotion recognition machine learning algorithm **224**. The user device **120** (e.g., via the emotion recognition machine learning algorithm **224**) may extract a first set of facial features **310a** from the first image **226a** and/or the sensor data **228a**. The first set of facial features **310a** may include eye tracking features, a posture of the face of the user **102**, and shapes of different portions of the face of the user **102**, among other information about physical attributes of the face of the user **102** shown in the image **226a** and/or derived from the sensor data **228a**.

(45) The user device **120** (e.g., via the emotion recognition machine learning algorithm **224**) may determine that the user **102** exhibited a first emotion **156a** in the first image **226a** based on the first set of facial features **310a**. For example, the emotion recognition machine learning algorithm **224** may be pre-trained to determine the emotion **156a** of the user **102** from the image **226a** based on a training dataset that comprises a set of image samples, each labeled with a respective emotion **156** and/or a set of sensor data samples, each labeled with a respective emotion **156**, similar to that described in FIG. 2.

(46) The user device **120** may determine that the user **102** selects the avatar **104a** for use within the virtual environment **130**. For example, the user **102** may be presented with a list of avatars **104a** on a display of the user device **120** and customize a particular avatar **104a** per their current mood (e.g., emotion **156a**). In customizing the avatar **104a**, the user **102** may select different attributes of the avatar **104a**, such as cloth, face, etc. In response, the user device **120** may determine that the emotion **156a** of the user **102** may be associated to the avatar **104a**. The user device **120** may then associate or link the emotion **156a** of the user **102** to the avatar **104a**. The user device **120** may perform these operations on multiple occasions during the training phase **302** to build up the training dataset **154a**. Therefore, the training dataset **154a** may gradually grow as new instances of user's emotions **156** and respective avatars **104** are detected and determined.

(47) The user device **120** may then use the training dataset **154a** to train the avatar prediction machine learning algorithm **150**. For example, the training dataset **154a** may be fed to the avatar prediction machine learning algorithm **150**. The avatar prediction machine learning algorithm **150** may learn to associate each user emotion data **156** to the respective avatar(s) **104**. For example, the avatar prediction machine learning algorithm **150** may extract a set of features from a user emotion data **156** (that indicates the emotion of the user **102**, such as happy, sad, etc.). The extracted set of features may be represented by a vector. The avatar prediction machine learning algorithm **150** may then learn to associate the extracted set of features to the respective avatar(s) **104a**.

(48) In certain embodiments, the avatar prediction machine learning algorithm **150** may reside in the memory of the user device **120**. Thus, in certain embodiments, the operations with respect to predicting the avatars **104** based on the user's emotion **156** and presenting the recommended avatar(s) **104** to the user **102** may be performed by the user device **120**.

(49) In certain embodiments, the avatar prediction machine learning algorithm **150** may reside in the memory of the virtual environment management device **140**. Thus, in certain embodiments, the user device **120** may communicate the determined emotion **156a** to the virtual environment management device **140** in messages (**106** in FIG. 1) for processing. The virtual environment

management device **140** may also determine that the user **102** selects the avatars **104a** by monitoring the activities of the members of the virtual environment **130**.

(50) The virtual environment management device **140** (e.g., via the avatar prediction machine learning algorithm **150**) may perform similar operations to that described above to build the training dataset **154a** and to train the avatar prediction machine learning algorithm **150**.

(51) Testing Phase to Test the Avatar Prediction Machine Learning Algorithm

(52) In the testing phase **304**, assume that the user **102** wants to operate an avatar **104** within the virtual environment **130**. In response, similar to that described above in the training phase **302**, the user **102** may log into their profile and access the platform to enter the virtual environment **130**.

The camera **208** may capture a second image **226b** of the user **102**. The camera **208** may communicate the captured image **226b** to the processor of the user device **120**. The user device **120** may receive the second image **226b** from the camera **208**. Receiving the second image **226b** from the camera **208** may be in response to the user **102** being logged into their user profile with respect to the virtual environment **130** to access the virtual environment **130** at a second timestamp. The second timestamp may be later than the first timestamp when the first image **226** was received or captured. Similarly, the user device **120** may receive the sensor data **228b** from the biometric device **218**. For example, the sensor data **228b** may include the average of the potential voltage signals between the front and back of the eye of the user **102**, eye movements, eye tracking information, among other information during the time window that the user **102** has put on the user device **120**.

(53) The user device **120** may feed the image **226b** and the sensor data **228b** to the emotion recognition machine learning algorithm **224**. The user device **120** (e.g., via the emotion recognition machine learning algorithm **224**) may extract a second set of facial features **310b** from the second image **226b** and/or the sensor data **228b**. The second set of facial features **310b** may include eye tracking features, the posture of the face of the user **102**, and shapes of different portions of the face of the user **102**, among other information about physical attributes of the face of the user **102** shown in the image **226b** and/or derived from the sensor data **228b**.

(54) The user device **120** (e.g., via the emotion recognition machine learning algorithm **224**) may determine that the user **102** exhibited a second emotion **156b** in the second image **226b** based on the second set of facial features **310b**. For example, the emotion recognition machine learning algorithm **224** may be pre-trained to determine the emotion **156b** of the user **102** from the image **226b** based on a training dataset that comprises a set of image samples, each labeled with a respective emotion **156** and/or a set of sensor data samples, each labeled with a respective emotion **156**, similar to that described in FIG. 2.

(55) In response to detecting the emotion **156b** of the user **102**, the user device **120** may feed the emotion data **156b** to the avatar prediction machine learning algorithm **150**. The user device **120** (e.g., via the avatar prediction machine learning algorithm **150**) may predict one or more avatars **104** based on the historical avatar selections of the user **102** when they had the same emotion as the current emotion **156b**. In this process, the user device **120** (e.g., via the avatar prediction machine learning algorithm **150**) may compare the first set of facial features **310a** to each entry of the training dataset **154a**. For example, the user device **120** (e.g., via the avatar prediction machine learning algorithm **150**) may compare the first set of facial features **310a** to the second set of facial features **310b**. The user device **120** (e.g., via the avatar prediction machine learning algorithm **150**) may determine whether more than a threshold percentage (e.g., 90%, 95%, etc.) of the first set of facial features **310a** correspond to the counterpart facial features from among the second set of facial features **310b**. In response to determining that more than the threshold percentage of the first set of facial features **310a** correspond to the counterpart facial features from among the second set of facial features **310b**, the user device **120** (e.g., via the avatar prediction machine learning algorithm **150**) may determine that the first emotion **156a** of the user **102** (shown in the first image **226a**) corresponds to the second emotion **156b** of the user **102** (shown in the second image **226b**).

(56) In certain embodiments, the user device **120** (e.g., via the avatar prediction machine learning algorithm **150**) may determine that the second emotion **156b** is the same or similar to the first emotion **156a**. For example, second emotion **156b** may be classified as a related emotion, and the first emotion **156a** may be classified as a related, serene, calm, or contented emotion. The first set of facial features **310a** may be represented by a first facial feature vector **312a** that comprises a first set of numerical values. The second set of facial features **310b** may be represented by a second facial feature vector **312b** that comprise a second set of numerical values. Thus, in certain embodiments, comparing the first set of facial features **310a** to the counterpart facial features from among the second set of facial features **310b** may be in response to comparing the first facial feature vector **312a** to the second facial feature vector **312b**.

(57) In certain embodiments, determining that the user **102** exhibits the first emotion **156a** in the second image **226b** may be in response to determining that more than a threshold percentage (e.g., 90%, 95%, etc.) of the first set of numerical values of the first facial feature vector **312a** corresponds to the counterpart numerical values of the second facial feature vector **312b**. In response, the user device **120** may present one or more recommended avatars **104a** that each is labeled with the first emotion **156a** for user selection, for example, on the application platform where the user **102** can access and view to select one of the avatars **104a**. In this manner, the user device **120** (e.g., via the avatar prediction machine learning algorithm **150**) may look into the training dataset **154a** and predict the recommended avatars **104a** for the user **102** based on their emotion **156b**. The user **102** may select one of the recommended avatars **104a**. The user device **120** may implement the selected avatar **104a** on behalf of the user **102** so that user **102** is able to operate the selected avatar **104a** in the virtual environment **130**.

(58) In certain embodiments, the avatar prediction machine learning algorithm **150** may reside in the memory of the virtual environment management device **140**. Thus, in certain embodiments, the user device **120** may communicate the determined emotion **156b** to the virtual environment management device **140** in messages (**106** in FIG. **1**) for processing. In certain embodiments, the operations with respect to the avatar prediction machine learning algorithm **150** may be performed by the virtual environment management device **140**. For example, in response to receiving the determined emotion data **156b**, the virtual environment management device **140** (e.g., via the avatar prediction machine learning algorithm **150**) may predict the recommended avatar(s) **104a** based on the training dataset **154a**, similar to that described above. The avatar prediction machine learning algorithm **150** (e.g., via the avatar prediction machine learning algorithm **150**) may then present the recommended avatar(s) **104a** to the user **102**.

(59) Continuing the testing phase **304**, in response to determining that more than the threshold percentage of the first set of extracted facial features **310a** corresponds to the counterpart facial feature from among the second set of extracted facial features **310b**, the user device **120** may present the first avatar **104a** and second avatar **104b** that are labeled with the first emotion **156a** to the user for user selection. The user **102** may select one of the presented avatars **104a, b**. The user device **120** may determine whether the user **102** selects one of the first and second avatars **104a, b**. In response to determining that the user **102** selects one of the first and second avatars **104a, b**, the user device **120** may implement the selected avatar **104a, b**, on behalf of the user **102** so that user **102** can navigate through the virtual environment **103** using the selected avatar **104a, b**.

(60) In some cases, the user **102** may select another avatar **104**, e.g., manually. If it is determined that the user **102** selects another avatar **104** (e.g., a third avatar **104c**) that is different from the first and second avatars **104a, b**, the user device **120** may associate the third avatar **104c** to the first emotion **156a** and update the training dataset **154a** to include the third avatar **104c** labeled with the first emotion **156a**. In this manner, the training dataset **154a** may grow and be updated based on new selections of the user **102**. The updated training dataset **154a** may be used for further avatar prediction operations.

(61) As described above, in certain embodiments, the user device **120** may build up the training

dataset **154a** on multiple occasions when the user **102** wants to operate an avatar in the virtual environment **130**. For example, the user device **120** may receive a third image **226c** from the camera **208** when the user **102** logs into their profile at the virtual environment **130**. The user device **120** may also receive sensor data **228c** from the biometric device **218**. The sensor data **228c** may include the average of the potential voltage signals between the front and back of the eye of the user **102**, eye movements, eye tracking information, among other information during the time window that the user **102** puts on the user device **120**.

(62) The user device **120** may extract a third set of facial features **310c** from the third image **226c** and/or the sensor data **228c**, similar to that described above via the emotion prediction machine learning algorithm **224**. In response, the user device **120** may determine the emotion of the user **102** as shown in the image **226c** and/or derived from the sensor data **228c**. The user device **120** may compare the determined emotion to each of the emotions **156** previously classified in the training dataset **154a**. For example, assume that it is determined that the user **102** exhibits the first emotion **156a** in the third image **226c** based on the third set of facial features **310c** and comparing with the training dataset **42a**. The user device **120** may determine that the user **102** selects a second avatar **104b** for use in the virtual environment **130**. In response, the user device **120** may associate the first emotion **156a** to the second avatar **104b**. The user device **120** may add the second avatar **104b** that is labeled with the first emotion **156a** to the training dataset **154a**.

(63) In certain embodiments, the operations with respect to updating the training dataset **154** may be performed by one or both of the user device **120** and the virtual environment management device **140**. For example, the user device **120** may communicate the determined emotion **156a** of the user shown in the image **226c** and/or derived from the sensor data **228c** and the selected avatar **104b** to the virtual environment management device **140**. The virtual environment management device **140** may associate the selected avatar **104b** to the detected emotion **156a** and add the second avatar **104b** that is labeled with the first emotion **156a** to the training dataset **154a**.

(64) Although certain operations are described to be performed by the user device **120**, in certain embodiments, any of the operations of the user device **120** may be performed by the virtual environment management device **140**, such as determining the emotion of the user, determining that an avatar is selected, associating the determined emotion **156a** to a selected avatar **104a**, building the training dataset **154a**, predicting the recommended avatars **104a**, presenting the recommended avatars **104a** to the user **102**, among others. Similarly, although certain operations are described to be performed by the virtual environment management device **140**, in certain embodiments, any of the operations of the virtual environment management device **140** may be performed by the user device **120**, such as determining the emotion of the user, determining that an avatar is selected, associating the determined emotion **156a** to a selected avatar **104a**, building the training dataset **154a**, predicting the recommended avatars **104a**, presenting the recommended avatars **104a** to the user **102**, among others.

(65) The system **100** of FIG. **1** may be configured to perform similar operations to predict one or more virtual locations **132** for the user **102** to visit based on the determined emotion **156** of the user **102** from the images **226a-c** and/or sensor data **228a-c**, by implementing the virtual location prediction machine learning algorithm **152** and the training dataset **154b**, similar to that described above with respect to predicting one or more avatars **104a** based on the emotion **156** of the user **102**. In response, the user device **120** and/or the virtual environment management device **140** may provide one or more recommended virtual locations **132** to the user **102** by communicating the recommended virtual locations **132** in a message to the interface application platform that the user **102** uses to access the virtual environment **130**. The recommended virtual locations **132** may be the same or similar to the virtual locations **132** that the user **102** previously visited when they had the same or similar emotion **156** as the currently determined emotion **156** for the user **102**.

(66) Method for Predicting Avatars Based on User's Emotions

(67) FIG. **4** illustrates an example flowchart of a method **400** for predicting avatars **104** based on

user's emotions **156**. Modifications, additions, or omissions may be made to method **400**. Method **400** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the system **100**, user device **120**, virtual environment management device **140**, or components of any of thereof performing operations, any suitable system or components of the system may perform one or more operations of the method **400**. For example, one or more operations of method **400** may be implemented, at least in part, in the form of software instructions **148**, **222** of FIGS. **1** and **2**, respectively, stored on non-transitory, tangible, machine-readable media (e.g., memory **146**, **220** of FIGS. **1** and **2**, respectively) that when run by one or more processors (e.g., processor **142**, **202** of FIGS. **1** and **2**, respectively) may cause the one or more processors to perform operations **402-434**.

(68) At operation **402**, the user device **120** receives the first image **226a** from the camera **208**. For example, the camera **208** may capture the first image **226a** of the user **102** when the user **102** logs into their profile at the virtual environment **130** using their credentials, e.g., in response to the user request or automatically.

(69) At operation **404**, the user device **120** extracts a first set of facial features **310a** from the first image **226a**. For example, the user device **120** may implement the emotion recognition machine learning algorithm **224** to extract the first set of facial features **310a** from the first image **226a**, similar to that described in FIGS. **2** and **3**.

(70) At operation **406**, the user device **120** determines that the user **102** exhibited a first emotion **156a** in the first image **226a** based at least on the first set of facial features **310a**. In certain embodiments, the user device **120** may also determine the first emotion **156a** based on the sensor data **228a**, similar to that described in FIG. **3**.

(71) At operation **408**, the user device **120** may determine that the user **102** selects a first avatar **104a** for use within the virtual environment **130**. At operation **410**, the user device **120** may associate the first emotion **156a** of the user **102** with the first avatar **104a**.

(72) At operation **412**, the user device **120** may add the first emotion **156a** of the user **102** labeled with the first avatar **104a** to the training dataset **154a**. The user device **120** may perform the operations **402-412** on multiple times during the training phase **302** to grow the training dataset **154a**, similar to that described in FIG. **3**. In certain embodiments, the operations of building and growing the training dataset **154a** (e.g., operations **402-412**) may be performed by the virtual environment management device **140**.

(73) At operation **414**, the user device **120** receives a second image **226b** from the camera **208**. For example, on another occasion that the user **102** wants to access the virtual environment **130**, the user **102** may log into their profile at the virtual environment **130** using their credentials, similar to that described in FIG. **3**.

(74) At operation **416**, the user device **120** may extract a second set of facial features **310b** from the second image **226b**, e.g., by implementing the emotion recognition machine learning algorithm **224**, similar to that described in FIGS. **2** and **3**. At operation **418**, the user device **120** determines that the user **102** exhibits a second emotion **156b** in the second image **226b** based at least on the second set of facial features **310b**. In certain embodiments, the user device **120** may also determine the second emotion **156b** based on the sensor data **228b**, similar to that described in FIG. **2**.

(75) At operation **420**, the user device **120** may compare each of the first set of facial features **310a** with a counterpart facial feature from among the second set of facial features **310b**. In this operation, the first feature vector **312a** may be compared to the second feature vector **312b**. For example, the Euclidean distance between the first and second feature vectors **312a** and **312b** may be determined. In certain embodiments, the user device **120** may perform the operation **420**, e.g., by implementing the avatar prediction machine learning algorithm **150**.

(76) At operation **422**, the user device **120** may determine whether more than a threshold percentage of the first set of facial features **310a** corresponds to the counterpart facial features from among the second set of facial features **310b**. In this process, the user device **120** may determine

whether the Euclidean distance between the first and second feature vectors **312a** and **312b** is less than a threshold distance (e.g., 0.1, 0.2, etc.). If it is determined that the Euclidean distance between the first and second feature vectors **312a** and **312b** is less than the threshold distance, it may be determined that the more than the threshold percentage of the first set of facial features **312a** corresponds to the counterpart facial features from among the second set of facial features **310b**. Otherwise, it may be determined that the less than the threshold percentage of the first set of facial features **312a** corresponds to the counterpart facial features from among the second set of facial features **310b**. If it is determined that the more than the threshold percentage of the first set of facial features **312a** corresponds to the counterpart facial features from among the second set of facial features **310b**, method **400** may proceed to operation **430**. Otherwise, method **400** may proceed to operation **424**.

(77) At operation **424**, the user device **120** may determine that the first emotion **156a** of the user **102** does not correspond to the second emotion **156b** of the user **102**. At operation **426**, the user device **120** may determine that the user **102** selects a second avatar **104b** for use within the virtual environment **130**. At operation **428**, the user device **120** adds the second emotion **156b** labeled with the second avatar **104b** to the training dataset **154a**.

(78) At operation **430**, the user device **120** may determine that the first emotion **156a** of the user **102** corresponds to the second emotion **156b** of the user **102**. At operation **432**, the user device **120** may present at least the first avatar **104a** for the user selection. At operation **432**, the user device **120** may implement the first avatar **104a** on behalf of the user **102** to operate within the virtual environment **130**.

(79) In certain embodiments, the virtual environment management device **140** may perform at least one of the operations **402-432**, e.g., by implementing the avatar prediction machine learning algorithm **150** and other software instructions. For example, any combination of the emotion data **156a, b**, sensor data **228a, b, c**, images **226a, b, c** may be provided to the virtual environment management device **140** and the virtual environment management device **140** may predict the one or more avatars **104** for the user **102** based on the emotion **156** of the user and analyzing historical activities and selections of the user **102** and the training dataset **154a**. Therefore, in certain embodiments, the virtual environment management device **140** may present a list of recommended avatars **104** to the user **102** for user selection and implement the selected avatar **104** on behalf of the user **102** to operate within the virtual environment **130**.

(80) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.

(81) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(82) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112 (f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system for predicting avatars for a user based on an emotion of the user, the system comprising: a camera configured to capture a set of images of a user, wherein the set of images comprises a first image and a second image; a processor operably coupled to the camera, and configured to: receive the first image from the camera; extract a first set of facial features from the first image; determine that the user exhibited a first emotion in the first image based at least in part upon the first set of extracted facial features; determine that the user selects a first avatar for use within a virtual environment; associate the first emotion of the user to the first avatar; build a training dataset that comprises the first emotion of the user labeled with the first avatar; receive the second image from the camera; extract a second set of facial features from the second image; determine that the user exhibited a second emotion in the second image based at least in part upon the second set of extracted facial features; compare at least one of the first set of extracted facial features to a counterpart facial feature from among the second set of extracted facial features; determine that more than a threshold percentage of the first set of extracted facial features correspond to the counterpart facial features from among the second set of extracted facial features; in response to determining that more than the threshold percentage of the first set of extracted facial features correspond to the counterpart facial features from among the second set of extracted facial features: determine that the first emotion of the user corresponds to the second emotion of the user exhibited in the second image; present at least the first avatar that is labeled with the first emotion for user selection; and implement the first avatar on behalf of the user in the virtual environment.
2. The system of claim 1, wherein: receiving the first image from the camera is in response to the user being logged into a user profile to access the virtual environment at a first timestamp; and receiving the second image from the camera is in response to the user being logged into the user profile to access the virtual environment at a second timestamp.
3. The system of claim 1, wherein: the first set of facial features is represented by a first facial feature vector that comprises a first set of numerical values; the second set of facial features is represented by a second facial feature vector that comprises a second set of numerical values; and comparing at least one of the first set of extracted facial features to the counterpart facial features from among the second set of extracted facial features is in response to comparing the first facial feature vector to the second facial feature vector.
4. The system of claim 3, wherein determining that the user exhibits the first emotion in the second image is in response to determining that more than the threshold percentage of the first set of numerical values of the first facial feature vector corresponds to counterpart numerical values of the second facial feature vector.
5. The system of claim 1, wherein the processor is further configured to: receive a third image from the camera; extract a third set of facial features from the third image; determine that the user exhibited the first emotion in the third image based at least in part upon the third set of extracted facial features; determine that the user selects a second avatar for use within the virtual environment; associate the first emotion of the user to the second avatar; and add the second avatar labeled with the first emotion of the user to the training dataset.
6. The system of claim 5, wherein the processor is further configured to: in response to determining that more than the threshold percentage of the first set of extracted facial features correspond to the counterpart facial feature from among the second set of extracted facial features: present the first avatar and the second avatar that are labeled with the first emotion for user selection; determine whether the user selected one of the first avatar and the second avatar; and in response to determining that the user selects one of the first avatar and the second avatar, implement the selected avatar on behalf of the user in the virtual environment.
7. The system of claim 6, wherein the processor is further configured to: in response to determining

that the user selects a third avatar that is different from the first avatar and the second avatar; associate the third avatar to the first emotion; and update the training dataset to include the third avatar labeled with the first emotion.

8. A method for predicting avatars for a user based on an emotion of the user, the method comprising: capturing a set of images of a user, wherein the set of images comprises a first image and a second image; receiving the first image from a camera; extracting a first set of facial features from the first image; determining that the user exhibited a first emotion in the first image based at least in part upon the first set of extracted facial features; determining that the user selects a first avatar for use within a virtual environment; associating the first emotion of the user to the first avatar; building a training dataset that comprises the first emotion of the user labeled with the first avatar; receiving the second image from the camera; extracting a second set of facial features from the second image; determining that the user exhibited a second emotion in the second image based at least in part upon the second set of extracted facial features; comparing at least one of the first set of extracted facial features to a counterpart facial feature from among the second set of extracted facial features; determining that more than a threshold percentage of the first set of extracted facial features correspond to the counterpart facial feature from among the second set of extracted facial features; in response to determining that more than the threshold percentage of the first set of extracted facial features correspond to the counterpart facial feature from among the second set of extracted facial features: determining that the first emotion of the user corresponds to the second emotion of the user exhibited in the second image; presenting at least the first avatar that is labeled with the first emotion for user selection; and implementing the first avatar on behalf of the user in the virtual environment.

9. The method of claim 8, wherein: receiving the first image from the camera is in response to the user being logged into a user profile to access the virtual environment at a first timestamp; and receiving the second image from the camera is in response to the user being logged into the user profile to access the virtual environment at a second timestamp.

10. The method of claim 8, wherein: the first set of facial features is represented by a first facial feature vector that comprises a first set of numerical values; the second set of facial features is represented by a second facial feature vector that comprises a second set of numerical values; and comparing at least one of the first set of extracted facial features to the counterpart facial features from among the second set of extracted facial features is in response to comparing the first facial feature vector to the second facial feature vector.

11. The method of claim 10, wherein determining that the user exhibits the first emotion in the second image is in response to determining that more than the threshold percentage of the first set of numerical values of the first facial feature vector corresponds to counterpart numerical values of the second facial feature vector.

12. The method of claim 8, further comprising: receiving a third image from the camera; extracting a third set of facial features from the third image; determining that the user exhibited the first emotion in the third image based at least in part upon the third set of extracted facial features; determining that the user selects a second avatar for use within the virtual environment; associating the first emotion of the user to the second avatar; and adding the second avatar labeled with the first emotion of the user to the training dataset.

13. The method of claim 12, further comprising: in response to determining that more than the threshold percentage of the first set of extracted facial features correspond to the counterpart facial features from among the second set of extracted facial features: present the first avatar and the second avatar that are labeled with the first emotion for user selection; determine whether the user selected one of the first avatar and the second avatar; and in response to determining that the user selects one of the first avatar and the second avatar, implement the selected avatar on behalf of the user in the virtual environment.

14. The method of claim 13, further comprising: in response to determining that the user selects a

third avatar that is different from the first avatar and the second avatar: associating the third avatar to the first emotion; and updating the training dataset to include the third avatar labeled with the first emotion.

15. A non-transitory computer-readable medium storing instructions that when executed by a processor, cause the processor to: receive a first image of a user from a camera; extract a first set of facial features from the first image; determine that the user exhibited a first emotion in the first image based at least in part upon the first set of extracted facial features; determine that the user selects a first avatar for use within a virtual environment; associate the first emotion of the user to the first avatar; build a training dataset that comprises the first emotion of the user labeled with the first avatar; receive a second image of the user from the camera; extract a second set of facial features from the second image; determine that the user exhibited a second emotion in the second image based at least in part upon the second set of extracted facial features; compare at least one of the first set of extracted facial features to a counterpart facial features from among the second set of extracted facial features; determine that more than a threshold percentage of the first set of extracted facial features correspond to the counterpart facial features from among the second set of extracted facial features; in response to determining that more than the threshold percentage of the first set of extracted facial features correspond to the counterpart facial features from among the second set of extracted facial features: determine that the first emotion of the user corresponds to the second emotion of the user exhibited in the second image; present at least the first avatar that is labeled with the first emotion for user selection; and implement the first avatar on behalf of the user in the virtual environment.

16. The non-transitory computer-readable medium of claim 15, wherein: receiving the first image from the camera is in response to the user being logged into a user profile to access the virtual environment at a first timestamp; and receiving the second image from the camera is in response to the user being logged into the user profile to access the virtual environment at a second timestamp.

17. The non-transitory computer-readable medium of claim 15, wherein: the first set of facial features is represented by a first facial feature vector that comprises a first set of numerical values; the second set of facial features is represented by a second facial feature vector that comprises a second set of numerical values; and comparing at least one of the first set of extracted facial features to the counterpart facial features from among the second set of extracted facial features is in response to comparing the first facial feature vector to the second facial feature vector.

18. The non-transitory computer-readable medium of claim 17, wherein determining that the user exhibits the first emotion in the second image is in response to determining that more than the threshold percentage of the first set of numerical values of the first facial feature vector corresponds to counterpart numerical values of the second facial feature vector.

19. The non-transitory computer-readable medium of claim 15, wherein the processor is further configured to: receive a third image from the camera; extract a third set of facial features from the third image; determine that the user exhibited the first emotion in the third image based at least in part upon the third set of extracted facial features; determine that the user selects a second avatar for use within the virtual environment; associate the first emotion of the user to the second avatar; and add the second avatar labeled with the first emotion of the user to the training dataset.

20. The non-transitory computer-readable medium of claim 19, wherein the instructions further cause the processor to: in response to determining that more than the threshold percentage of the first set of extracted facial features correspond to the counterpart facial feature from among the second set of extracted facial features: present the first avatar and the second avatar that are labeled with the first emotion for user selection; determine whether the user selected one of the first avatar and the second avatar; and in response to determining that the user selects one of the first avatar and the second avatar, implement the selected avatar on behalf of the user in the virtual environment.
